



12V BATTERY CHARGER-Cum-VARIABLE POWER SUPPLY

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The circuit presented here can charge a 12V lead-acid battery of 50Ah to 80Ah (even up to 100Ah) capacity and can even be used as up to 18V DC variable power supply of maximum 5A capacity, which is useful for a test bench. The circuit can automatically detect the presence of a battery connection and start charging. At that time, it disconnects the output provided as variable power supply. It also detects the wrong/reverse polarity connection of the battery terminals and raises an alarm. The charger initially charges the battery at a higher voltage (about 14.2V), and once it is fully charged, it maintains the battery charge at a constant voltage (about 13.4V).

Circuit and working

Circuit of the 12V battery charger-cum-variable power supply is shown in Fig. 1. It is built around two LM138 variable voltage regulators (IC1 and IC2), dual op-amp LM358 (IC3), 12V voltage regulator 7812 (IC4), two relays (with normal 12V, 1C/O PCB-mounted and 12V, 1C/O, 10A contact-current rating) and a couple of transistors.

LM138 is a 3-pin, 5A positive-voltage regulator available in TO-220 or TO-3 package. But continuous current flow of 5A generates a high temperature, which shuts down output of LM138 automatically due to its internal thermal protection.

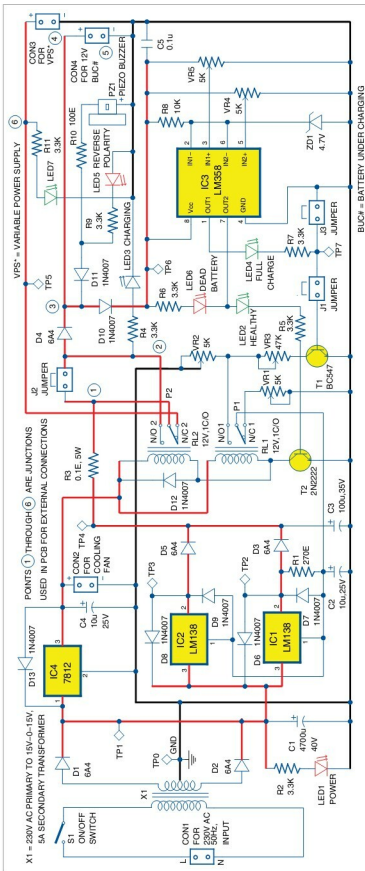


Fig. 1: Circuit of the 12V battery charger-cum-variable power supply